

Final project proposal

● Graded

Group

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 [View or edit group](#)

Total Points

3 / 3 pts

Question 1

Project Title

0.5 / 0.5 pts

✓ - 0 pts Correct

Question 2

Dataset

0.5 / 0.5 pts

2.1 Name and link to dataset

0.5 / 0.5 pts

✓ - 0 pts Correct

- 0.5 pts Link does not work

2.2 If using a different dataset than what we have provided, please provide a URL to a webpage or citation that describes what the dataset contains (e.g. how it was collected, dataset size, etc.) and write 3-4 sentences on why this dataset will make for an interesting final project. If using a dataset we provided, skip this section.

0 / 0 pts

- 0 pts Climate dataset

✓ - 0 pts Outside dataset (approved)

- 0 pts Outside dataset (not approved, must switch to climate dataset)

💬 Sam recommends that you ...take a look at <https://distill.pub/2018/building-blocks/> for inspiration

Question 3

Brief writeup of what you intend to do (5-10 lines)

1 / 1 pt

✓ - 0 pts Correct

Question 4

5-6 static visualizations on your dataset

1 / 1 pt

✓ - 0 pts Correct

- 0.5 pts Between 1-4 static visualizations

- 1 pt No static visualizations

Project title

The Hidden Vocabulary of GPT-2: Visualizing Sparse Autoencoder Features from Neuronpedia

Team member

Krish Desai Nidhi Manayilakath Jingwen Gu

Dataset

We will visualize public sparse autoencoder (SAE) feature data from Neuronpedia, an open platform for mechanistic interpretability research. Our processed sample uses **GPT-2 small** features from residual stream SAE layers 2, 6, and 10, with 3,000 rows and 18 columns. Each row is one SAE feature and includes feature ID, layer, feature index, explanation, activation density, top logits, max activation, concept category, legibility score, and 2D coordinates. The data is publicly available, non-synthetic, and satisfies the requirement of at least 100 rows and 5 columns.

Source links:

- Neuronpedia: <https://www.neuronpedia.org/>
- API/docs: <https://docs.neuronpedia.org/api>
- Public exports:
<https://neuronpedia-datasets.s3.us-east-1.amazonaws.com/index.html?prefix=v1/>
- Local processed file: `data/processed/features.csv`

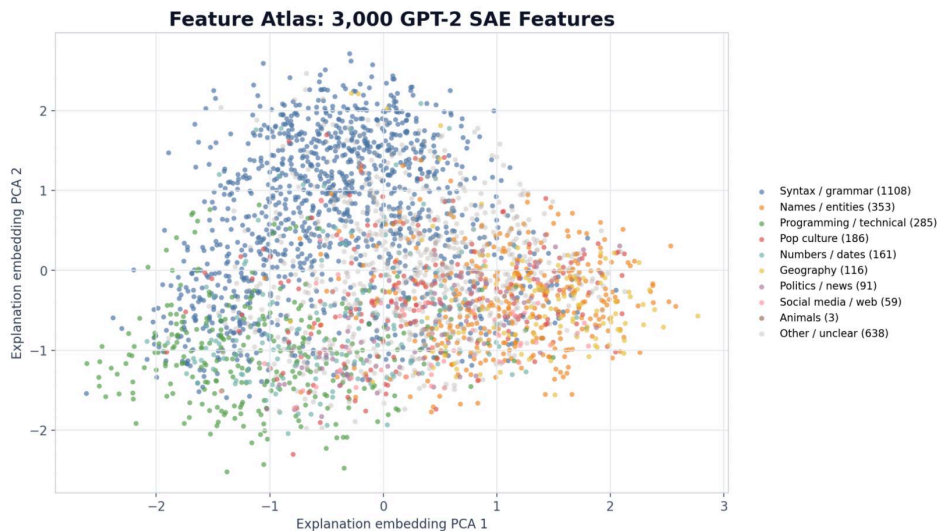
This dataset is interesting because it lets us visualize a language model's internal features rather than only its final outputs. Some SAE features have recognizable explanations, such as names, syntax, websites, money, technical terms, or public figures. Mapping thousands of these features can show how interpretable concepts cluster, differ across layers, and vary in clarity. The project also lets us discuss the limits of interpretability, since feature explanations can be useful but imperfect.

Brief Writeup Of What We Intend To Do

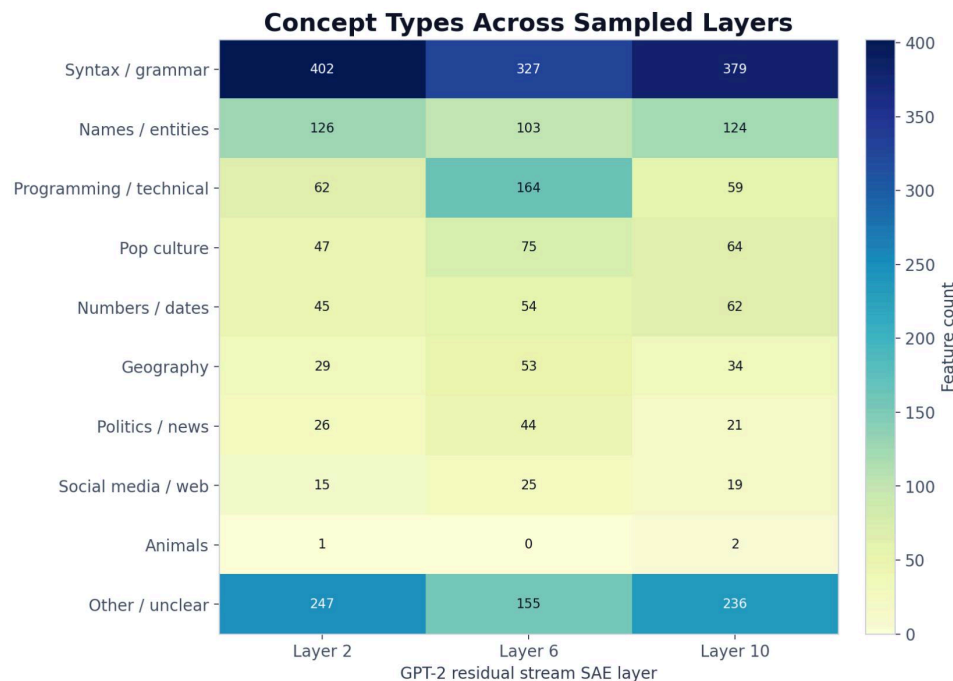
We will build an interactive explorable article about GPT-2's hidden feature space. Each point in the main atlas will represent one SAE feature. Users will be able to search, filter by layer or category, and click features for details. The project will compare feature categories across layers using heatmaps and summary charts. It will also show activation density and a simple legibility score for feature explanations. The goal is to make mechanistic interpretability data understandable to non-experts.

Static Visualizations On The Dataset

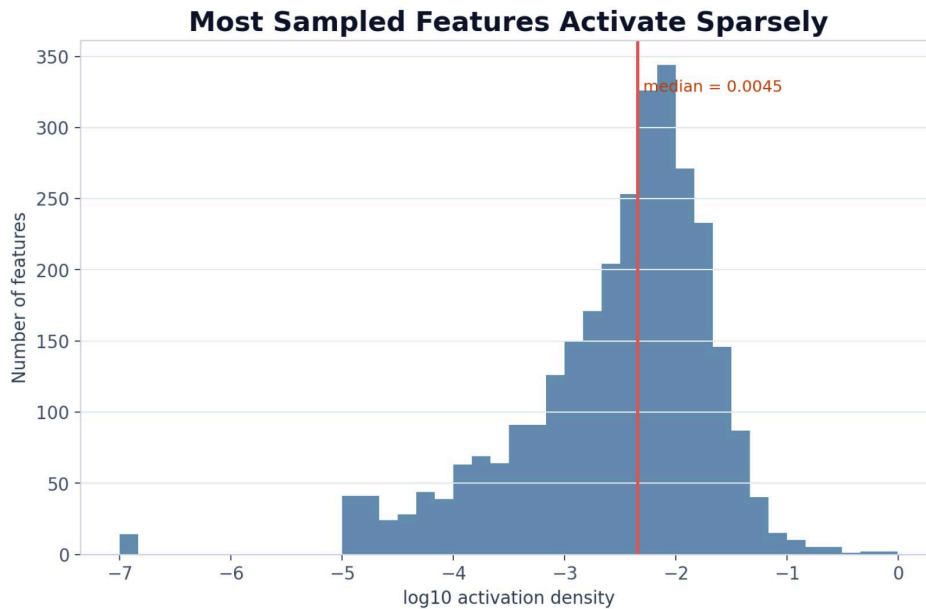
1. Feature Atlas Scatterplot: Shows all 3,000 features as points, colored by concept category.



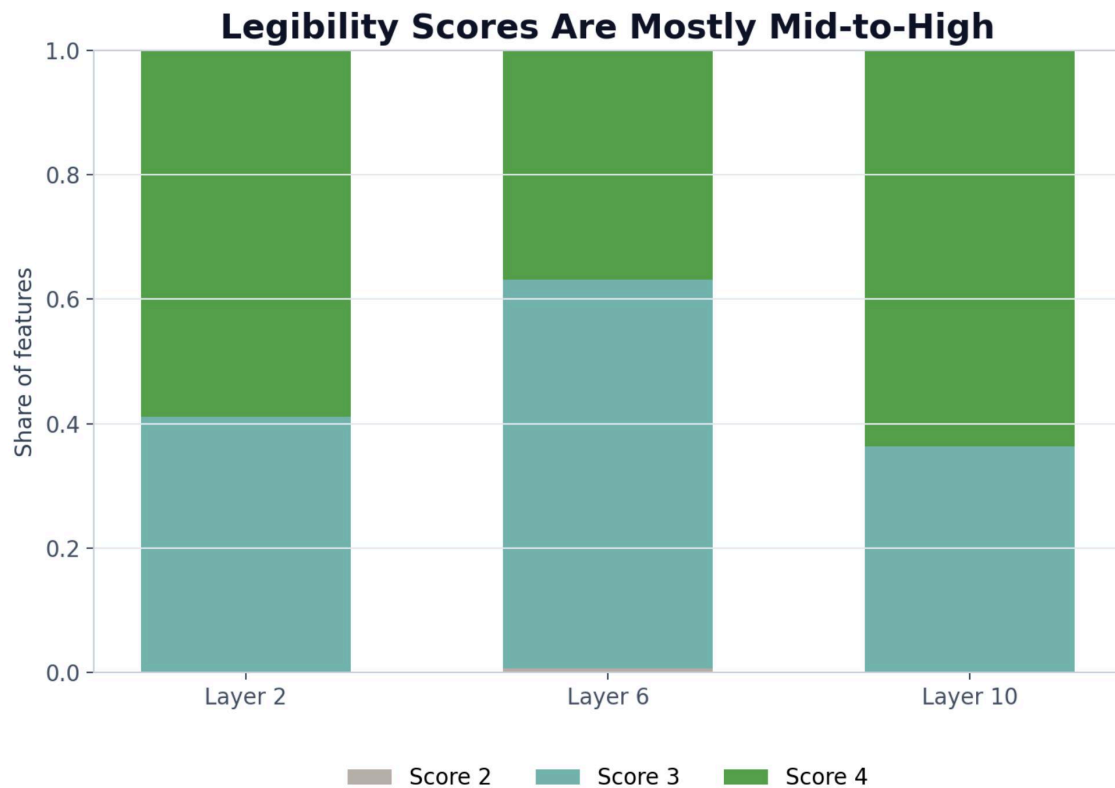
2. Layer-Category Heatmap: Compares concept category counts across layers 2, 6, and 10.



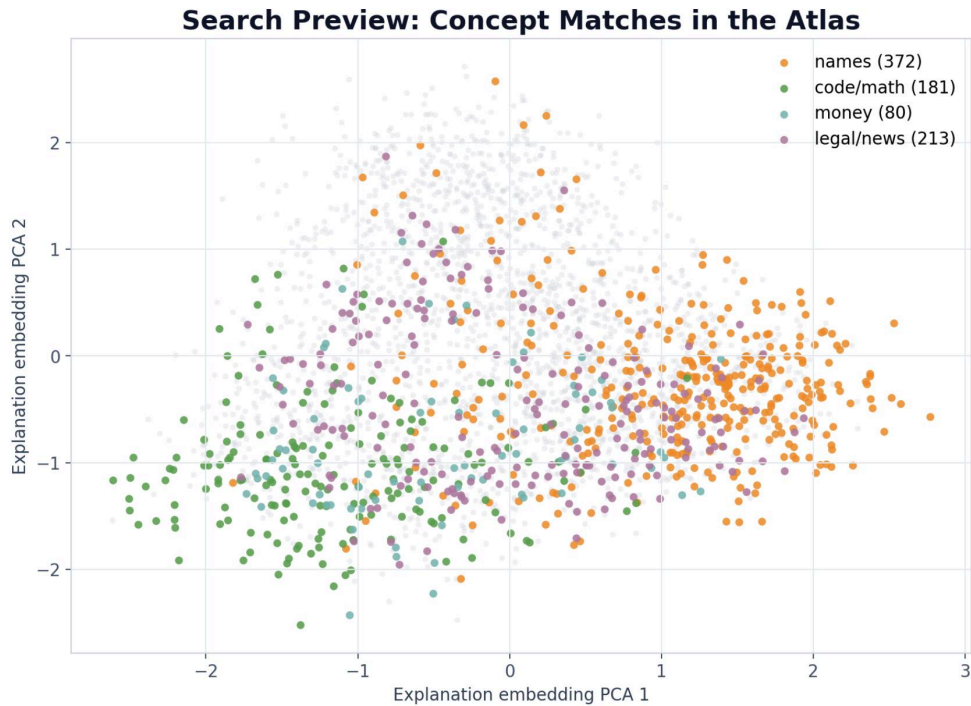
3. Search Highlight Atlas: Previews how search terms highlight matching features in the atlas.



4. Activation Density Histogram: Shows how sparse the sampled SAE features are.



5. Legibility Score By Layer: Compares explanation legibility across sampled layers.



6. Feature Detail Card: Shows the planned click-detail view for one selected feature.

Example Feature Detail Card

gpt2-small/2-res_scefr-ajt/110

Explanation mentions of the name "Frank."

Layer 2

Activation density 0.00773

Legibility score 4 / 5

Top positive logits fur, furt, Frank, lins, fort, olini, enstein, Geh

Top negative logits lLA, wrapper, disabilities, avers, td, weeney, arre, ignty

No questions assigned to the following page.

